

# Portugal Administrative GIS & Freguesia Population Dataset

## Project Overview

The project visualizes official geographic boundaries in Portugal both The Nomenclature of Territorial Units for Statistics (NUTS) and local administrative level (LAU) and combined it with population data at the Freguesia level. The final outputs are prepared as GIS-ready datasets and professional cartographic products.

The project integrates:

- NUTS 1, NUTS 2 and NUTS 3 statistical regions
- Local administrative level boundaries
- Freguesia-level population data

## Project Objectives

1. Building a clear administrative and statistical geographic hierarchy for Portugal.
2. Preparing NUTS 1, NUTS 2 and NUTS 3 datasets.
3. Processing local administrative level boundaries.
4. Integrating Freguesia population data and boundaries.
5. Performing spatial and attribute-level QA/QC.
6. Producing clean and deliverable GIS-ready Shapefile.
7. Generating professional maps for visualization and documentation.

## Data Processing Workflow

1. Downloaded NUTS and LAU datasets from Eurostat GISCO at Europe continental level.
2. Filtered the NUTS and LAU datasets for Portugal
3. Separated NUTS 1, nuts 2 AND nuts 3 Layers
4. Merged LAU layers of different region in Portugal
5. Downloaded the INE Population dataset. 6. Inspected geographic identifiers and attribute fields.
7. Matched population records to Freguesia geographic identifiers
8. Calculated and verified geographic attributes
9. Aggregated the population data in Freguesia level
10. Join population data to the Freguesia boundary
11. Created population-based symbology.
12. Performed QA/QC validation.
13. Exported final deliverable GIS datasets
14. Produced cartographic layouts and documentation

## Tools

**QGIS:** Select features, Merge vector layers, Aggregate, Join, Symbology, Attribute table, Vector geometry, Check validity, QA/QC, Layout

**Microsoft Word:** Documentation

**Microsoft Excel:** Data Dictionary

## Data Sources

Geographic level datasets were obtained from official Portuguese geographic data sources with reference year 2024. Population data with reference year 2021 were integrated at the freguesia level boundaries. Therefore, 2021 population represented using 2024 Freguesia administrative boundaries

The datasets with geographic reference are listed below:

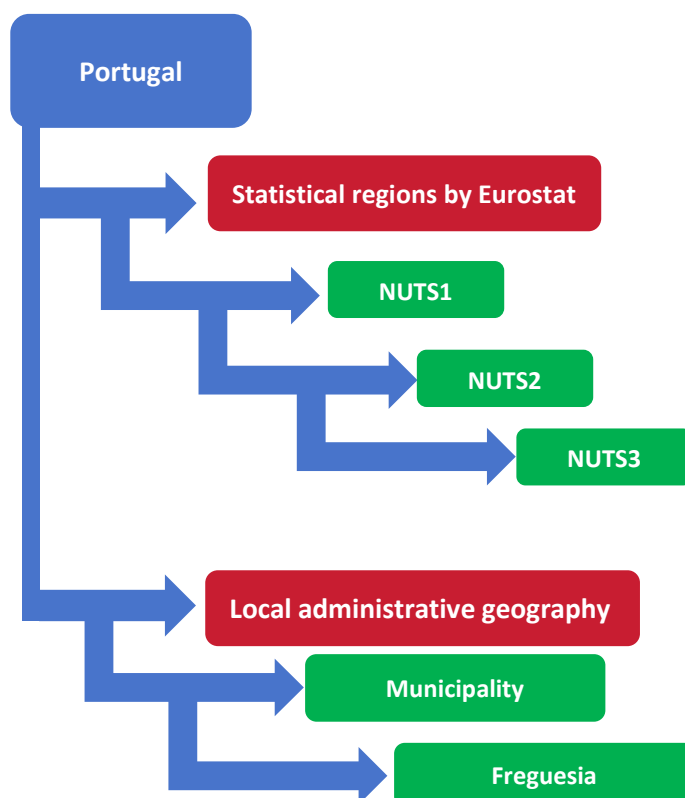
| Datasets                                          | Geographic level           | Usage                     |
|---------------------------------------------------|----------------------------|---------------------------|
| Eurostat GISCO NUTS 2024 <sup>1</sup>             | NUTS 1–3                   | Regional hierarchy        |
| Eurostat GISCO — LAU 2024 <sup>2</sup>            | Municipality and Freguesia | Administrative boundaries |
| Population dataset - INE Census 2021 <sup>3</sup> | Freguesia                  | Population attributes     |

LAU and NUTS: Year 2024, 1:1 Million, EPSG:3035 — ETRS89 / LAEA Europe  
Map Projection Grid: EPSG:4326 (WGS 84)

<sup>1</sup> <https://ec.europa.eu/eurostat/web/gisco/geodata/statistical-units/local-administrative-units>  
<sup>2</sup> <https://ec.europa.eu/eurostat/web/gisco/geodata/statistical-units/territorial-units-statistics>  
<sup>3</sup> <https://mapas.ine.pt/download/index2021LugaresFregs.phtml>

## Geographic Hierarchy

Portugal's geographic structure includes both territorial and local administrative units . The NUTS classification is used for statistical and regional level, while municipalities and Freguesias represent local administrative level geography.



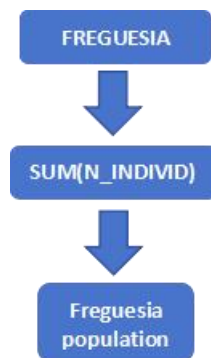
Geographic levels used in this project are listed below:

| Level                                                                                                                   | Geography                                          | Role                             |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------|
| Country                                                                                                                 | Portugal                                           | National level                   |
| NUTS 1                                                                                                                  | Territory of the Mainland and Autonomous Regions   | Statistical classification       |
| NUTS 2                                                                                                                  | Administrative Regions                             | Statistical and planning         |
| NUTS 3                                                                                                                  | Inter-municipal Communities and Metropolitan Areas | Detailed regional statistics     |
| Municipality                                                                                                            | Município                                          | Local administrative level       |
| Freguesia                                                                                                               | Freguesia                                          | Local administrative subdivision |
| NUTS 1, 2 and 3 are statistical subdivisions by Eurostat<br>Municipality and Freguesia are local administrative levels. |                                                    |                                  |

## Final Freguesia Population Dataset

Population data were joined to the freguesia administrative boundaries using geographic identifiers. Population data contained multiple records associated with the same freguesia were aggregated with sum function. Then, the relevant population values were being joined to the freguesia boundary layer. The final population field represents the 2021 population associated with the corresponding freguesia.

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The identifiers of attribute table are listed below:

| Field      | Description                     |
|------------|---------------------------------|
| GEO_ID     | Freguesia geographic identifier |
| FREGUESIA  | Freguesia name                  |
| MUNICIPIO  | Municipality                    |
| DISTRICT   | District                        |
| NUTS1      | NUTS 1 region                   |
| NUTS2      | NUTS 2 region                   |
| NUTS3      | NUTS 3 region                   |
| AREA_HA    | Area in hectares                |
| POPULATION | Total number of inhabitants     |
| COUNTRY    | Country                         |
| SOURCE     | Boundary/source information     |

## QA/QC Results

After performing Quality control the following result has been found:

| Check                    | Result |
|--------------------------|--------|
| Total Freguesia features | 3,259  |
| Missing GEO_ID           | 0      |
| Duplicate GEO_ID         | 0      |
| Invalid geometries       | 0      |
| Geometry errors          | 0      |
| Population NULL          | 302    |

Population coverage is incomplete because the population data source does not use exactly the same Freguesia geographic configuration as the administrative boundaries. As a result, 302 current Freguesia records could not be directly matched to a population value. These records were retained with NULL population values rather than assigning potentially incorrect values.

## Limitations

- Administrative boundaries and population data have different reference years.
- Some freguesias do not have corresponding population records in the population datasets.